

## BASARA GNANASARASWATHI CAMPUS KAKATIYA HILLS

**Sec: Star SC**

**Group-14**

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### SINGLE ANSWER TYPE

- Which of the following statements is incorrect ?  
 (a) Graphite is thermodynamically stable allotropic form of carbon  
 (b) Graphite formation from Diamond is endothermic reaction  
 (c) Distance between two layers in Graphite is 340 pm  
 (d) In Diamond each carbon atom is in  $sp^3$  hybridisation
- Which of the following on heating with conc.  $H_2SO_4$  at  $100^\circ C$  produces two neutral oxides?  
 (a) Malonic acid      (b) Urea      (c) Formic acid      (d) Calcium carbonate
- The element which is affected by water is  
 (a) C      (b) Sn      (c) Ge      (d) Si
- Which of the following has  $Si_2O_7^{6-}$  unit?  
 (a) Phenacite      (b) Thortveitite      (c) Beryl      (d) Spodumene
- Ultramarine is  
 (a) Phyllo-silicate      (b) Sorosilicate      (c) Amphibole      (d) Three-dimensional silicate
- The formula of sodalite is  
 (a)  $Na_8[(AlSiO_4)_6]S_2$       (b)  $Na_8[(AlSiO_4)_6]Cl_2$       (c)  $Na_8[(AlSiO_4)_6]SO_4$       (d)  $Na_8[(AlSiO_4)_6]CO_3$
- The compound 'A' is formed by dehydrating malonic acid with  $P_4O_{10}$  the number of sigma sp-sp bonds in 'A' is  
 (a) 4      (b) 1      (c) 2      (d) 3
- Which of the following pairs have different structures ?  
 (a)  $NF_3, NH_3$       (b)  $PF_3, PH_3$       (c)  $N(CH_3)_3, N(SiH_3)_3$       (d)  $SiF_4, CF_4$
- Silica dissolves in aqueous hydrofluoric acid and forms stable silicon complex. The hybridisation of silicon in the complex is  
 (a)  $sp^3d^2$       (b)  $sp^3$       (c)  $d^2sp^3$       (d)  $sp^3d$
- $SiF_4 + H_2O \rightarrow A \xrightarrow{1000^\circ C} B \xrightarrow{Na_2CO_3} C$  Identity B and C ?  
 (a)  $H_4SiO_4, Na_2SiO_3$       (b)  $SiO_2, Na_2S$       (c)  $SiO_2, Na_2CO_3$       (d)  $SiO_2, Na_2SiO_3$
- Which of the following statements is correct ?  
 (I) 'CO' gas is absorbed by  $FeSO_4$  solution

- (II)  $\text{CO}_2$  gas is absorbed by aqueous suspension of  $\text{Cu}_2\text{Cl}_2$
- (III)  $\text{CO}_2$  gas is absorbed by  $\text{KOH}$  solution
- (IV) 'CO' gas absorbed by aqueous suspension of  $\text{Cu}_2\text{Cl}_2$
- (a) I, II                      (b) III, IV                      (c) II, III                      (d) I, IV
12. The carbide which gives Methyl acetylene on hydrolysis is
- (a)  $\text{CaC}_2$                       (b)  $\text{Mg}_2\text{C}_3$                       (c)  $\text{Al}_4\text{C}_3$                       (d)  $\text{SiC}$
13. Decreasing order of 's' character in the following
- (I)  $\text{CO}_2$                       (II)  $\text{SiO}_2$                       (III) Graphite
- (a)  $\text{I} > \text{III} > \text{II}$                       (b)  $\text{II} > \text{III} > \text{I}$                       (c)  $\text{I} > \text{II} > \text{III}$                       (d)  $\text{III} > \text{II} > \text{I}$
14. A.  $\text{CCl}_4$  does not act as lewis acid  
 B. Silanes are strong reducing agents  
 C. Crystalline silica will have Diamond like structure  
 The correct answer is
- a. A&B are true                      b. Only 'A' is true                      c. Only 'b' is true                      d. All are true.
15. How many O-atoms are shared per  $\text{SiO}_4$  tetrahedral in silicate anion of Beryl mineral ?
- a) 4                      b) 3                      c) 2                      d) 1
16. Buckminster fullerene  $\text{C}_{70}$  has
- a) 25 hexagonal rings, 12 pentagonal rings    b) 20 hexagonal rings, 12 pentagonal rings  
 c) 25 pentagonal rings, 12 hexagonal rings    d) 20 pentagonal rings, 12 hexagonal rings
17. Carbogen is
- a) mixture of  $90-95\%\text{O}_2 + 5-10\%\text{CO}_2$                       b) used by pneumonia patients for respiration  
 c) used by victims of CO for respiration                      d) all of the above
18. The basic reaction involved in the synthesis of linear silicones is:
- (A) The hydrolysis of trimethyl chlorosilane  
 (B) The hydrolysis of diethyl dichlorosilane  
 (C) The hydrolysis of ethyl chlorosilane  
 (D) The acid hydrolysis of dimethyl silane
19. In which of the following silicates, only two corners per tetrahedron are shared ?
- i) Pyrosilicate                      ii) Cyclic silicate                      iii) Double chain silicate  
 iv) Single chain silicate                      v) 3D silicate                      vi) Sheet silicate
- 1) i, ii, iii                      2) iv and vi only                      3) i and vi only                      4) ii and iv only
20.  $\text{PbCl}_2$  is more ionic than  $\text{PbO}_2$  because
- 1) The radius of  $\text{Pb}^{2+}$  is more than that of  $\text{Pb}^{4+}$                       2) Of inert pair effect  
 3) Chlorine is more electronegative than oxygen  
 4) Chlorine atom is smaller than oxygen atom

21. In graphite which have several fused hexagonal rings of benzene the hybridisation state of each carbon atom and the bond order of each carbon - carbon bond are respectively  
 1)  $sp$ , 1.5                                  2)  $sp^2$ , 1.5                                  3)  $sp^2$ , 1.33                                  4)  $sp^3$ , 1.5
22. Carborundum on heating with caustic soda in the presence of air produces  
 1)  $Na_2SiO_3 + H_2$                                   2)  $Na_2SiO_3 + Na_2CO_3$                                   3)  $Na_2SiO_2 + H_2$                                   4)  $Na_2SiO_4 + O_2$

### MULTIPLE ANSWER TYPE

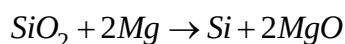
23. Which of the following statements are correct about silicones ?  
 A) They are formed by the hydrolysis of  $R_2SiCl_2$ .  
 B) They are polymers made up of  $R_2SiO_2$  units.  
 C) They are made up of  $SiO_4^{4-}$  units.                                  D) They are macro molecules.
24. Which of the following statements(s) is/are correct ?  
 A)  $PbI_4$  is not known because of the oxidising power of  $Pb(+IV)$  and the reducing power of  $I^-$   
 B)  $SnCl_2$  does not hydrolyse.                                  C)  $SnCl_2$  is reducing agent.  
 D) Tin (II) chloride is linear in shape.
25. Identify A, B, C and D in the following reaction sequence.
- $$1) Si + CH_3Cl \xrightarrow[570\text{ K}]{\text{Cu power}} (A) \xrightarrow{H_2O} (B)$$
- $\downarrow O_2$   
 (C)

Polymerisation  
 $\downarrow$
- $$(C) \xrightarrow[\text{electric arc}]{\text{Excess of coke}} (D) \text{ Silicone}$$

- A) The compound A is  $(CH_3)_2SiCl_2$     B) The compound C and D are  $SiO_2$  and  $SiC$  respectively  
 C) The compound B is  $(CH_3)_2Si(OH)_2$ .                                  D) The compound B is  $(CH_3)_2S=O_2$
26.  $SiO_2$  reacts with  
 A)  $Na_2CO_3$                                   B)  $CO_2$                                   C)  $HF$                                   D)  $HCl$
27. In which of the following all the atoms are in  $sp^3$  hybridization ?  
 A) Diamond                                  B) Carborundum                                  C) Crystalline silicon                                  D) Quartz
28. Which among the following statements are correct ?  
 A) Aqua dag and oil dag are made up of graphite  
 B) Graphite reacts with Conc.  $HNO_3$  to form mellitic acid  $C_6(COOH)_6$ .  
 C)  $C_3O_2$  is also toxic like  $CO$ .                                  D) Zircon,  $ZrSiO_4$  is a gemstone.
29. Which of the following statements is incorrect ?  
 A) The  $CO_2$  molecule behaves as a nonpolar molecule even though two of its resonating structures  
 $\bar{O} - C \equiv \bar{O}$  and  $\bar{O} \equiv C - \bar{O}$  are dipolar.  
 B) Carbon dioxide is the anhydride of the unstable dibasic acid  $O = C(OH)_2$ .  
 C) The  $CO_2$  molecule is linear because the carbon atom utilises its  $sp$  orbitals to form  $\sigma$  bonds.  
 D) The carbon atom is  $sp^2$  hybridised in the  $CO_2$  molecule as well as its hydrate  $H_2CO_3$ .

### PASSAGE-1

**Silicon.** The element silicon has two allotropic modification (a) Amorphous silicon (b) Adamantine silicon. A third allotropic modification has been observed and is of doubtful character known as graphitoidal silicon. Amorphous silicon is obtained by heating well-powdered quartz of finely silica with magnesium powder in a fireclay which is suitable for the manufacture of crucibles.



The product is washed with dilute  $HCl$  to wash away  $MgO$  and then with hydrofluoric acid to remove uncharged silica.

30. Crystalline silicon may be obtained by heating potassium silicon fluoride with  
(A) Al (B) C (C) Zn (D\*) Both (A) and (C)
31. Silicon reacts with fused aqueous NaOH to form  
(A)  $\text{SiO}_2$  (B)  $\text{SiO}_2^{2-}$  (C\*)  $\text{SiO}_3^{2-}$  (D)  $\text{SiO}$
32. When silica is heated with carbon then silica is reduced to  
(A) Si (B) SiO (C)  $\text{SiO}_3^{2-}$  (D\*) SiC

**PASSAGE-2** Rocks, clays and soils are made up of silicates of aluminium, iron, magnesium and other metals. All silicates are made up of  $\text{SiO}_4$  tetrahedral units in which Si is  $\text{sp}^3$  - hybridized is surrounded by four oxygen atoms. The  $\text{SiO}_4$  tetrahedra can be linked together in several different ways. Depending on the number of corners of the  $\text{SiO}_4$  tetrahedra shared, various kinds of silicates are formed

33. A fibrous mineral which can withstand red hot flames without any damage is  
1) Talc 2) Glass wool 3) Soap stone 4) Asbestos
34. Quartz watches contain  
1) A crystal of quartz as an essential component 2) A coating of quartz on the outer body  
3) Hands made of quartz 4) Silica coated on the number
35. Which of the following is not a crystalline form of silica ?  
1) Quartz 2) Tridymite 3) Cristobalite 4) Kieselguhr

### INTEGER ANSWER TYPE

36. How many oxygen atoms of  $[\text{SiO}_4]^{4-}$  are shared in three dimensional sheet silicate ?
37. C-60 consist both '5' membered and '6' membered rings. It contains  $x \times 2$  '5' membered rings. The value of x is \_\_\_\_\_.

### MATCHING TYPE

38. **Column – I**  
a)  $(\text{SiO}_3)_n^{2n-}$   
b) Silicone  
c)  $\text{Si}_2\text{O}_7^{6-}$   
d)  $(\text{Si}_4\text{O}_{11})_n^{6n-}$
- Column – II**  
p) Cyclic silicate  
q) Chain silicate  
r) Contain Si-O-Si bond  
s) Pyrosilicate
39. **Column – I**  
a) CO  
b)  $\text{PbO}_2$   
c) GeO  
d) SnO
- Column – II**  
p) Neutral  
q) Amphoteric  
r) Reducing agent  
s) Oxidising agent

### KEY

- |                                            |         |        |                                            |        |          |       |
|--------------------------------------------|---------|--------|--------------------------------------------|--------|----------|-------|
| 1. B                                       | 2. C    | 3. B   | 4. B                                       | 5. D   | 6. B     | 7. C  |
| 8. C                                       | 9. A    | 10. D  | 11. B                                      | 12. B  | 13. A    | 14. D |
| 15. C                                      | 16. A   | 17. D  | 18. B                                      | 19. 4  | 20. 1    | 21. 3 |
| 22. 2                                      | 23. ABD | 24. AC | 25. ABC                                    | 26. AC | 27. ABCD |       |
| 28. ABCD                                   | 29. ABC | 30. D  | 31. C                                      | 32. D  | 33. 4    | 34. 1 |
| 35. 4                                      | 36. 4   | 37. 6  | 38. a - p, q, r; b - r; c - r, s; d - q, r |        |          |       |
| 39. a - p, r; b - q, s; c - q, r; d - q, r |         |        |                                            |        |          |       |